

DATA ANALYTICS

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Building Energy Performance Through Machine Learning

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Abstract

This study investigates the impact of a set of constructive variables of the building envelope on heating and cooling demands across two distinct architectural shapes. The presented dataset, developed by the author for a doctoral thesis, leverages multi-objective optimization via the non-sorting genetic algorithm II (NSGA-II) to minimize both heating and cooling demand for residential buildings in Passo Fundo, southern Brazil, a Köppen Cfa climate classification, similar to Alicante, Spain. A computational effort totaling 480,000 simulations was conducted, varying construction and geometric parameters to yield 320 Pareto Front envelope solutions. These simulations were used as input for Random Forest models to predict annual energy consumption. Model interpretation utilized SHAP (SHapley Additive Explanations), enabling rigorous feature importance assessment for each building typology. The results show the indoor temperature setpoint as the most influential variable for both heating and cooling demand in both typologies. Among envelope parameters, roof thermal transmittance (Uroof) is the primary driver for heating in H buildings, while window-to-wall ratio (WWR) is most critical for cooling in Linear buildings. These findings point to the need to optimize envelope transmittance while considering the pivotal roles of building morphology and occupant behavior, which is reflected in setpoint choices.

Introduction

The construction sector represents a major share of global energy use, with building envelope thermal performance being a fundamental lever for achieving energy efficiency (IEA, 2022). Design choices that limit heat losses and manage solar gains have substantial potential to mitigate building carbon footprints and advance global sustainability goals (Ürge-Vorsatz et al., 2020). Despite advances in simulation-based energy assessments, there remains an acute gap in comparative research regarding how architectural morphologies (particularly shape and transparency ratios) condition the influence of envelope thermal transmittance on heating and cooling energy demand (Ascione et al., 2022; D'Agostino & Parker, 2018).

In the context of climate change and escalating energy demand, the design of energy-efficient residential buildings is especially challenging in hot-summer humid subtropical climates like Köppen Cfa, where peak cooling periods alternate with significant heating needs. Architects and engineers must balance multiple, often competing, design variables to minimize energy consumption while maintaining comfort and cost-effectiveness.

While optimization algorithms like NSGA-II identify Pareto-optimal solutions for energy performance, they do not inherently explain which architectural parameters most strongly influence heating and cooling requirements, nor how these variables interact. This gap constrains the translation of computational results to actionable guidance for design practice (Benincá et al., 2023).

Research Question: Which architectural design parameters most significantly impact heating and cooling energy demand in residential buildings under Cfa climate conditions?

Context and Dataset: H and Linear residential buildings in Passo Fundo, Brazil (Köppen Cfa). Are the result of 400,000 unique building configurations simulated (EnergyPlus) and optimized using NSGA-II (BESOS). The multi-objective optimization target was to minimize cooling (kWh/m²·year) and heating (kWh/m²·year) at same time using the variables were:

- Building shape (L or H);
- Window-to-Wall Ratio (WWR: 10%, 15%, 20%, 25%);
- Insulation thickness by wall orientation (north/south/east/west, rockwool in mm);

- U-value for each wall and roof ($\text{W}/\text{m}^2\cdot\text{K}$);
- Roof insulation thickness (rockwool in mm);
- Solar absorptance of walls/roof (color/factor);
- Glazing thickness (single glass, mm);
- Natural ventilation setpoint (AFN, $^{\circ}\text{C}$)

Main Objective: to build interpretable predictive models (Random Forest + SHAP) that quantify variable importance, capture non-linear effects, and provide actionable explanations for energy performance, bridging computational design and real-world decision-making.

Main Goals:

1. **Understand the Data:** exploratory analysis (EDA) to visualize distributions, correlations, and Pareto front solutions.
2. **Build Predictive Models:** machine learning (Random Forest/Polynomials) to predict heating/cooling demand from variables inputs.
3. **Derive Actionable Insights:** use SHAP to reveal parameter importance and interpret non-linear variable interactions.

Methodology

Datasets were generated using parametric building performance simulations encompassing geometric (shape, WWR) and physical (insulation thickness by orientation, roof, solar absorptance, glass thickness, ventilation setpoint) features. U-values were calculated based on optimized rockwool insulation thickness. A train-test split was used for Random Forest modeling in accordance with state-of-the-art practice (Zhao et al., 2021).

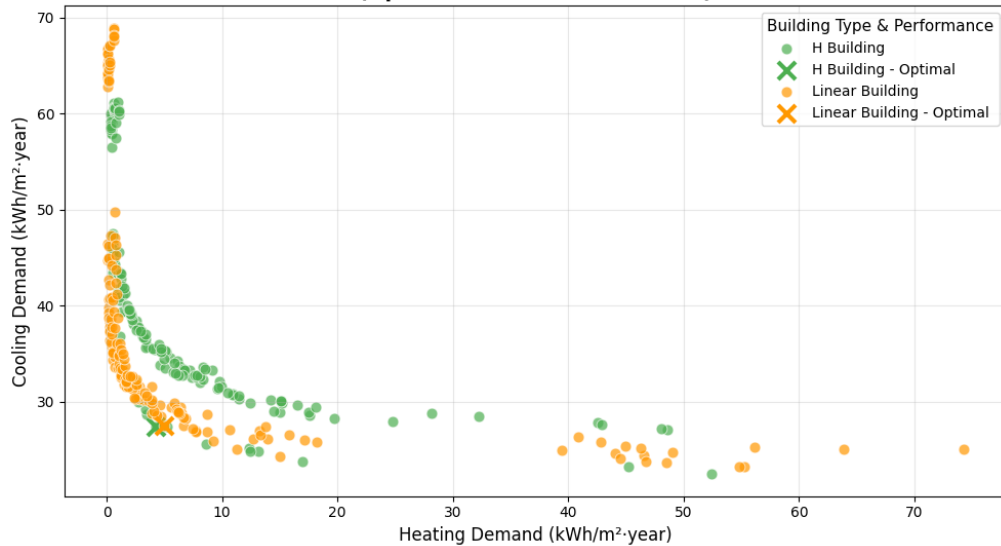
During multi-objective optimization, NSGA-II explored configurations from highly insulated facades ($U = 0.01 \text{ W}/\text{m}^2\text{K}$) to poorly insulated ($U = 2.60 \text{ W}/\text{m}^2\text{K}$), minimal and maximal glazing, and WWRs from 10–25%. These extremes, generated by the algorithm, are key design edge cases (not outliers) and were maintained for model generalizability. To handle these physically meaningful extremes, machine learning preprocessing used RobustScaler, which centers data on the median and interquartile range, preserving valuable variability without distortion due to skewed distributions (see Results). Extreme yet valid simulation cases (e.g., U values from 0.01 to 2.60 $\text{W}/\text{m}^2\text{K}$) were scaled using RobustScaler to retain boundary behaviors essential for robust machine learning generalization.

Also, it was conducting a comparison of Random Forest and Polynomials, where adding selected polynomial interactions improves predictive performance for heating demand but has a slight negative impact on cooling demand. For this matter, for the rest of the analysis it was chosen to use the Random Forest model

Model interpretation was performed with SHAP, decomposing predictions into feature-level attributions (Lundberg & Lee, 2017). Separate analyses for H and Linear buildings enabled comparative sensitivity assessment.

Results

**Heating vs Cooling Demand by Building Shape
(Optimal Points Marked with X)**

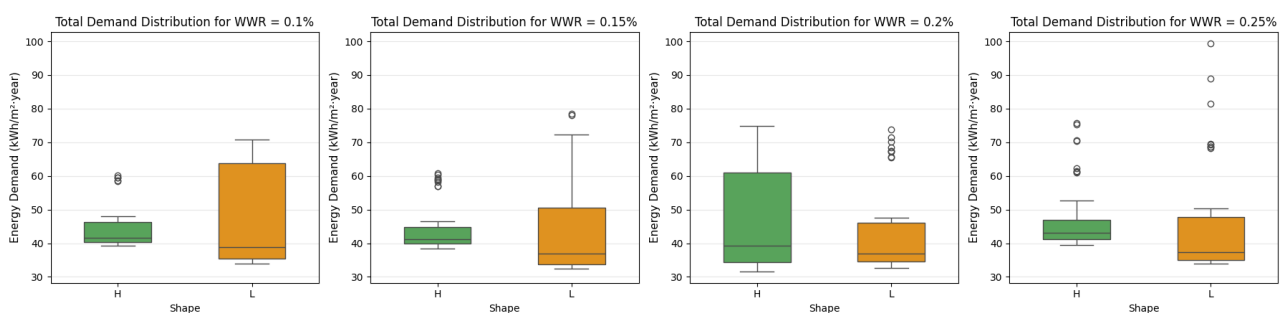


Y-axis (cooling) and X-axis (heating) are well defined and draw 3 clusters of results, for both buildings. The solutions that minimize heating are between 57 kWh/m².yr and 65 kWh/m².yr, solutions minimizing both demands among 45 kWh/m².yr (cooling) and until 15 kWh/m².yr (heating) and solutions minimizing cooling from 45 kWh/m².yr to 58 kWh/m².yr.

ID_Optimization	Gen	WWR	Shape	IT North	Unorth	IT South	Usouth	IT East	Ueast	...	Uwall	IT Roof	Uroof	SA Walls	SA Roof	IT Glazing	Setpoint	Cooling demand	Heating demand	Total demand
HW20_14	13.0	0.2	H	0.018	1.28	0.093	0.41	0.048	0.68	...	0.79	0.103	0.37	0.21	0.24	0.003	21.67	27.36	4.28	31.63

ID_Optimization	Gen	WWR	Shape	IT North	Unorth	IT South	Usouth	IT East	Ueast	...	Uwall	IT Roof	Uroof	SA Walls	SA Roof	IT Glazing	Setpoint	Cooling demand	Heating demand	Total demand
LW15_12	11.0	0.15	L	0.032	1.12	0.023	1.48	0.013	0.5	...	1.12	0.088	0.42	0.2	0.22	0.005	21.71	27.53	4.94	32.47

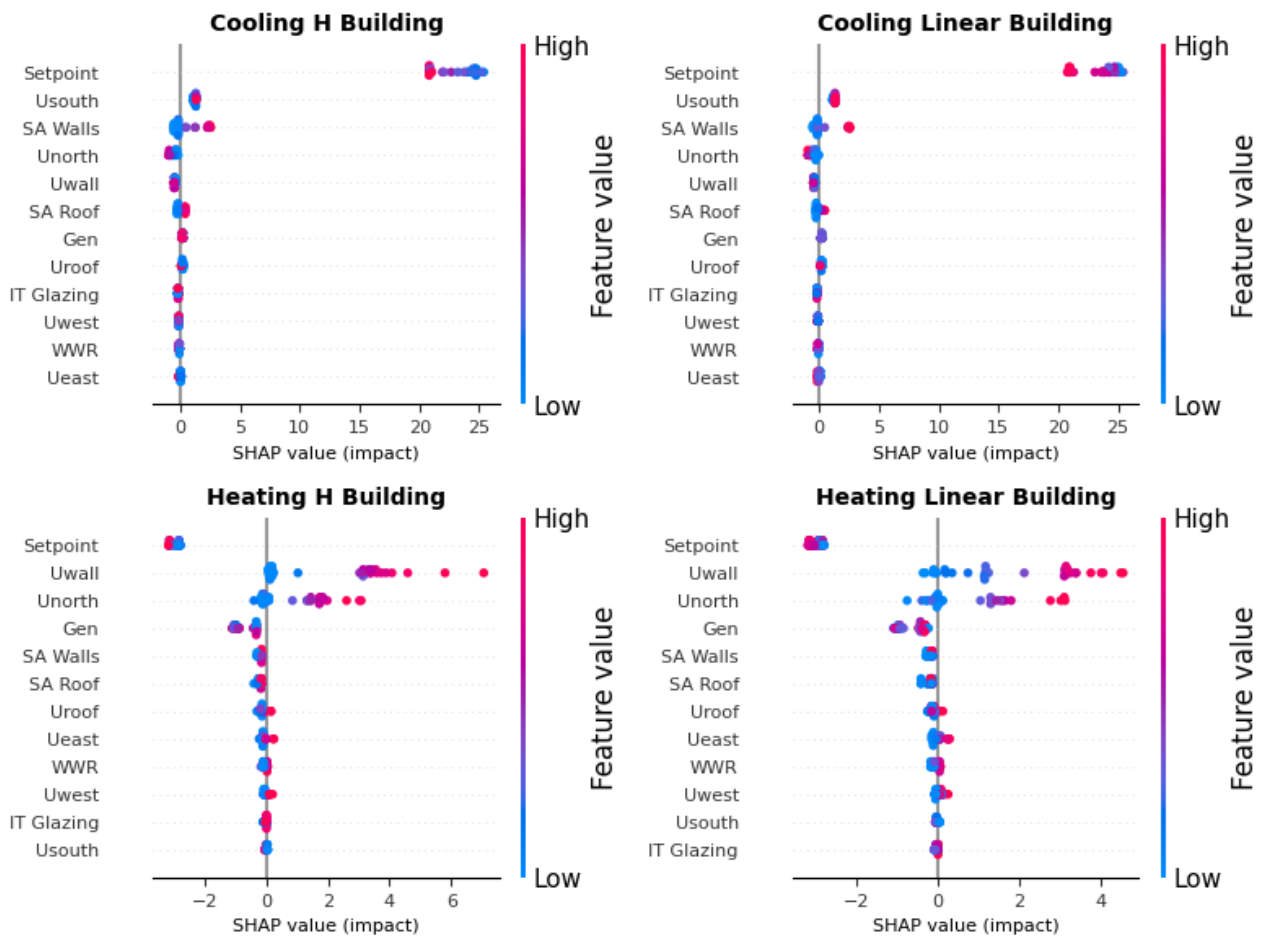
Comparison of Total Demand by WWR and Shape



H buildings consistently have lower total energy demand than Linear buildings across all WWR values, with the difference being most pronounced at 20 % WWR (this mismatch value is suspected to be some EnergyPlus simulation error during the multi-objective phase), where H building median demand is ~38 kWh/m².year vs. ~45 kWh/m².year for Linear building. At low WWRs (10%, 15%), Linear buildings exhibit higher variability and outliers, suggesting less consistent performance. As WWR increases, both shapes show rising demand, but H buildings maintain better control over energy use, likely due to more favorable solar exposure and compactness.

H buildings are more energy-efficient, especially at moderate to high WWRs. This supports the idea that building geometry significantly impacts energy performance, with H building offering better thermal balance and reduced demand.

SHAP Feature Importance H vs Linear Buildings



SHAP reveals that while Setpoint is universally important, building shape changes how physical parameters matter: H buildings are driven by envelope performance (insulation, absorptance). Linear buildings are driven by operational control (setpoint), with weaker dependence on insulation. This supports shape-specific design strategies, optimizing insulation in H buildings and focusing on thermal comfort control in Linear forms.

In this analysis, SHAP values are employed to uncover the building-shape-specific drivers of cooling and heating energy demand in H buildings and Linear buildings, revealing distinct behavioral patterns rooted in architectural form. For H buildings, cooling demand is predominantly influenced by operational setpoints and south-facing thermal transmittance (Usouth), indicating that solar exposure and indoor temperature settings are primary contributors to heat gain, while heating demand is overwhelmingly governed by wall insulation performance (Uwall), with secondary contributions from roof and north-facing envelope properties, underscoring the critical role of passive design strategies in mitigating winter heat loss. In contrast, Linear buildings exhibit a stronger dependence on operational parameters, with setpoint emerging

as the dominant factor for both cooling and heating demands, while physical envelope characteristics such as wall insulation (U_{wall}) and solar absorptance (SA Walls) play a secondary, still significant, role, suggesting that energy performance in Linear forms is more responsive to occupant behavior and control than to passive architectural interventions. These findings highlight that building geometry fundamentally reshapes the hierarchy of energy drivers: H buildings benefit most from optimizing envelope performance, whereas Linear buildings achieve greater efficiency through thermal setpoint management, reinforcing the necessity of shape-sensitive, context-aware design guidelines in energy-efficient architecture.

Discussion

Findings confirm that thermal transmittance is essential, but its true influence is governed both by morphology and the setpoint temperature. For H buildings, the large, exposed roof area makes Uroof critical for heating; in Linear forms, WWR becomes central for cooling. Yet, SHAP results reinforce the centrality of setpoint: it overrules physical parameters as the dominant driver for both heating and cooling loads.

This insight aligns with recent literature emphasizing occupant behavior and control as dominant in real-world building energy outcomes (Hong et al., 2020; O'Brien & Gunay, 2020). SHAP delivers quantitative, transparent guidance for prioritizing interventions. For true energy performance optimization, design should balance advanced envelope solutions with robust strategies for comfort and user behavior management.

Conclusion

This study illustrates that, while envelope transmittance optimization remains absolutely vital for building energy performance, its effectiveness—and even relevance—depends highly on building form and especially on occupant-controlled variables like setpoint temperature. In both H and Linear buildings, setpoint is the dominant variable. For heating in H buildings, Uroof retains a leading role; for cooling in Linear buildings, WWR is crucial. SHAP-based sensitivity analyses provide actionable, interpretable model insights supporting more efficient, morphology- and behavior-driven design strategies. Future research should extend this approach to varied climates, typologies, and occupancy schedules to further validate and generalize these results.